Release Notes

Ceph v20.2.4 with SSE-S3 KMIP

Base: Ceph v20.2.4 (Tentacle) + SSE-S3 KMIP feature + 9 backports
Build: 20.2.4-40-g28f95a921c81
Image: harbor.clyso.com/c-customers/gridscale/ceph/ceph:gridscale-v20.2.4-sse-s3-kmip-v1
Branch: https://github.com/clyso/ceph-dev/tree/release-sse-s3-kmip-v20.2.4
Previous release: gridscale-v20.2.1-sse-s3-kmip-v1

What Changed

- Rebased from Ceph v20.2.1 to v20.2.4, which includes four upstream security fixes (upstream release notes).
- The SSE-S3 KMIP feature is unchanged: no new configuration options and no change to how keys or encrypted objects are stored.
- 9 commits backported (listed below).

Upstream Pull Requests

- #69258: SSE-S3 for KMIP backend
- #72431: rgw/kmip: fail KMIP requests that cannot get a connection
- #72432: rgw/kmip: check SSL_CTX_new() and log credential errors
- #72452: rgw/kmip: add socket I/O timeout for KMIP connections
- #72456: rgw/kmip: add multithreading support to the KMIP client
- libkmip: ceph/libkmip, branch wip-kmip-ng

Downstream Backports

Nine upstream commits were backported to this release. Six of them fix build, unit test and lint failures found during the unit test run. The other three are upstream bug fixes for RGW and cephadm.

Commit	Why
rgw: copy_obj_data() record uncompressed size for a compressed source	Wrong object size after server-side copy of a compressed object (#75715, #75716)
cephadm: skip bluestore-label keyring update when only caps differ	OSD reconfig rotated the keyring and restarted the OSD even when the key was unchanged (#80440)
mgr/cephadm: check ok-to-stop before OSD redeploy/reconfig	cephadm restarted OSDs without checking that it was safe to stop them (#80440)
test/osd: add missing stringify.h include for TestOSDMap.cc	Build failure in make check
mon/MonClient: delete MonConnection move assignment	-Werror build failure
common/entity_name: print type names in get_valid_types_as_str()	ceph-conf cram test failure
test/cli: update cram expectations for cephx auth changes	run-cli-tests failure after the cephx CVE fix
qa/tasks: fix flake8 findings in ceph.py and radosbench.py	run-tox-qa lint failure
cephadm: black-format OSD key rotation code in daemons/ceph.py	cephadm format check failure

Testing

- **Unit tests:** run against the release branch; failures fixed by the backports above.
- **Teuthology:** several suites were run. The most relevant to this feature is rgw:crypt, which covers SSE-KMS and SSE-S3, including a PyKMIP-backed SSE-S3 job. Failures are listed under Known Issues.
- **Developer testing:** test_sse.py, an SSE-S3 smoke test against a PyKMIP server, covering PUT/GET, HEAD headers, multipart, range reads, overwrites, server-side copy and bucket default encryption. All 25 checks passed. Scripts: github.com/supriti/ceph-rgw-kmip-sse-s3

Known Issues

Found in CLYSO's Teuthology runs; all are present in upstream v20.2.4.

Tracker	Issue	Status
#80514	dedup: buffer overflow aborts radosgw (see below)	Not fixed in tentacle
#78431	Multisite: test_object_acl fails	Cause not identified
#72948	Multisite: intermittent failures in sync disable/enable	Fixed on master, not in tentacle
#78582	Multisite: test_bucket_log_trim_after_delete_bucket_secondary_reshard fails	Fixed in tentacle after v20.2.4
#78594	Lifecycle: test_lifecycle_deletemarker_expiration_with_days_tag fails	Not fixed

Do not run any radosgw-admin dedup command. The subcommands are still present although full dedup is disabled in tentacle. dedup estimate kills every radosgw daemon; dedup stats kills radosgw-admin. No data is at risk, and restarting the daemons restores service (#80514).

Operational Gaps and Risks

Unchanged from the previous release.

- **radosgw-admin bucket rm does not delete the bucket's KEK.** Use S3 DeleteBucket for encrypted buckets.
- **Orphan KEK if a destroy fails.** Rare; watch logs for "WARN: failed to destroy orphan KEK". No data impact.
- **KMIP server restart.** At most one failed S3 request (HTTP 500) per active worker; clients recover on retry.